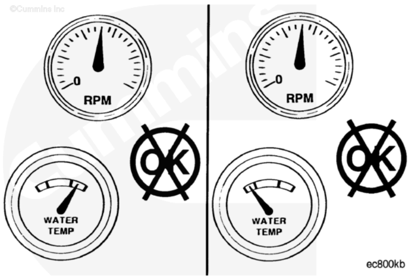
Trainings

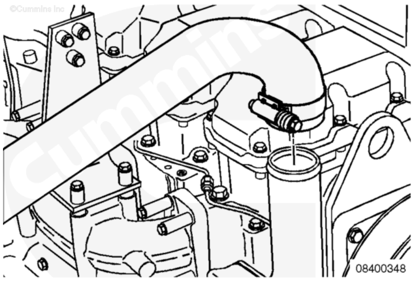
Leak Test

The engine thermostat and thermostat seal **must** operate properly in order for the engine to operate in the most efficient heat range. Overheating or overcooling will shorten engine life.



Complete this test with the engine coolant temperature below 50°C [120°F].

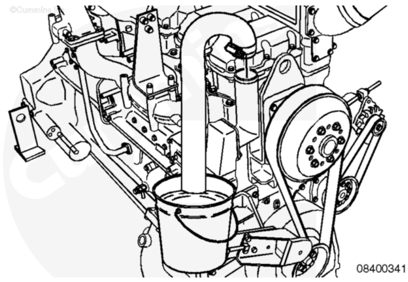
Remove the upper engine radiator hose from the thermostat housing.



Install a hose on the thermostat housing outlet long enough to reach a remote dry container.

Install and tighten the hose clamp on the housing outlet.

Put the end of the hose in the dry container.



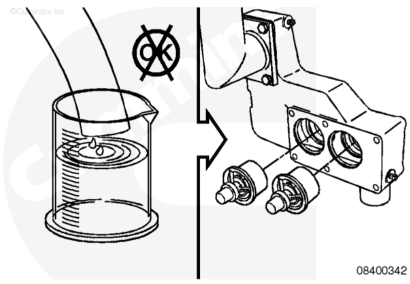
Operate engine at rated speed for 1 minute.

Shut the engine OFF and measure the amount of coolant collected.

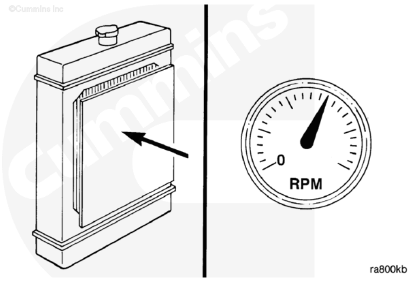
If more than 100 cc [3.3 fl oz] of coolant is collected, the thermostat or the thermostat seal is leaking.

Remove the thermostat and test operation as described in Procedure 008-013 (/qs3/pubsys2/xml/en/procedures/89/89-008-013.html).

Remove the test equipment and install the upper radiator hose.

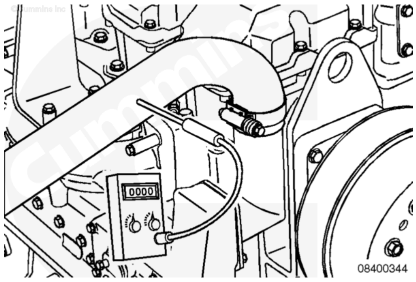


Restrict the radiator air flow. Operate the engine until the coolant temperature rises to 90-93°C [195-200°F].

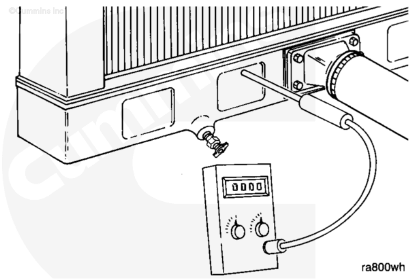


Record the temperature of the coolant outlet hose. An increase in temperature indicates the thermostats have started to open.

Use a contact pyrometer (shown) or install a temperature gauge in the desired location(s) prior to operating the engine.

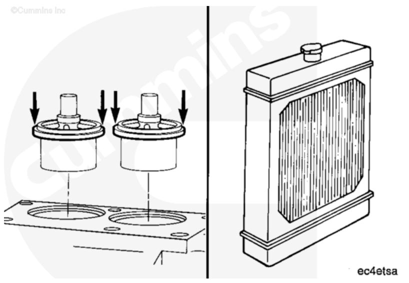


Record the temperature of the radiator bottom tank or coolant out tube.



If the difference in temperature is more than 8°C [15°F], either the thermostats are **not** fully open or the radiator core is plugged.

To replace the thermostats, refer to Procedure 008-013 (/qs3/pubsys2/xml/en/procedures/89/89-008-013.html). To check the radiator, refer to Procedure 008-042 (/qs3/pubsys2/xml/en/procedures/89/89-008-042.html).



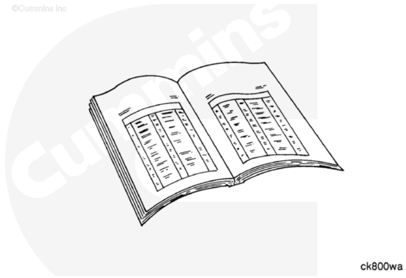
Preparatory Steps

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Headed coolant spray or steam can cause personal injury.



- Drain the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/89/89-008-018.html).
- Remove the hose from the thermostat housing. Refer to Procedure 008-045 (/qs3/pubsys2/xml/en/procedures/89/89-008-045.html).
- If equipped, remove the torque converter coolant hose.
- Disconnect the coolant temperature sensor on the side of the thermostat housing.

Remove

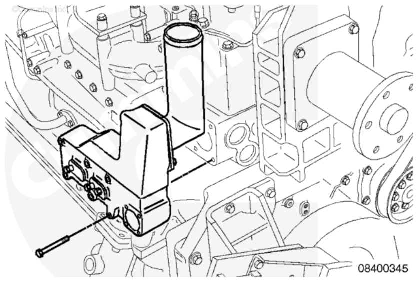
Remove the six long capscrews.

Remove the thermostat housing.

Remove and discard the gasket.

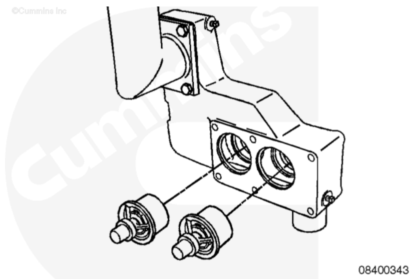
It may be necessary to loosen the bypass tube retainer and bypass tube to remove the thermostat housing.

Discard the bypass tube o-ring.



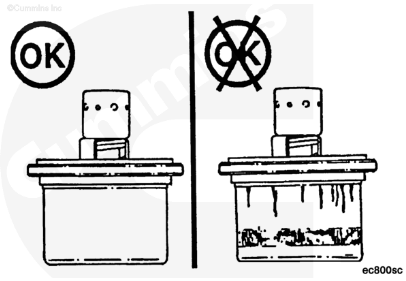
Remove the thermostats.

Remove the thermostat seals. Refer to Procedure 008-016 (/qs3/pubsys2/xml/en/procedures/89/89-008-016.html).



Inspect for Reuse

Inspect the thermostat for damage.

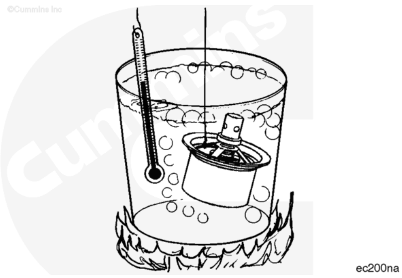


Test

Note : Do not allow the thermostat or thermometer to touch the container.

Suspend the thermostat and a 100°C [212°F] thermometer in a container of water.

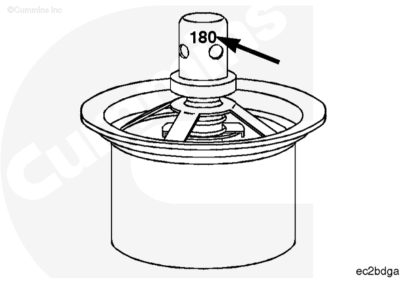
Heat the water and check the thermostat as follows.



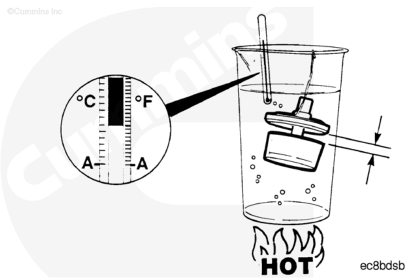
The nominal operating temperature is stamped on the thermostat.

- Thermostat **must** begin to open within 2°C [3°F] of nominal temperature.
- Thermostat **must** be fully open at 12°C [22°F] above nominal temperature.

The fully open distance between the thermostat flange and housing is 11.05 mm [0.43 in].



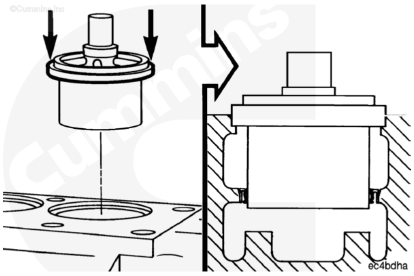
Remove the container from the heat. Check to see if the thermostat returns to the closed position.



Install

Install the thermostat seals. Refer to Procedure 008-016 (/qs3/pubsys2/xml/en/procedures/89/89-008-016.html).

Install the thermostat by pushing on the outer rim.

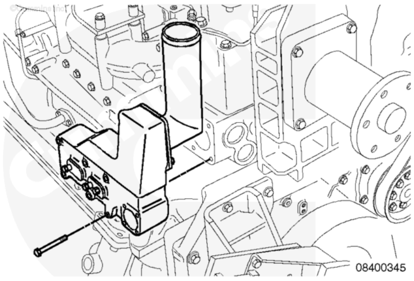


Install a new gasket.

Install the thermostat housing and capscrews.

Tighten the capscrews.

Torque Value: 66 n•m [50 ft-lb]

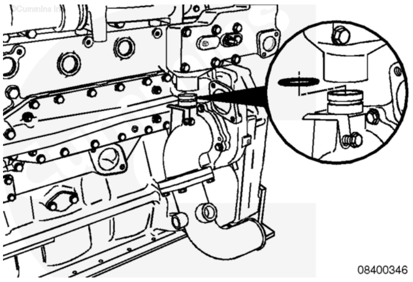


Install a new o-ring on the bypass tube. Use vegetable oil to lubricate the o-ring.

Slide the bypass tube and retainer into position.

Tighten the capscrew and the hose clamps.

Capscrew	31 n.m	[23 ft-lb]



Finishing Steps

- Connect the coolant temperature sensor on the side of the thermostat housing.
- Install the radiator hose.
- If equipped, install the torque converter coolant hose.
- Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/89/89-008-018.html).
- Operate the engine to 70°C [160°F] coolant temperature. Check for leaks.